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Excellence in Overcoming Well Integrity Challenges: Innovative Annulus Intervention Revives Wells at Risk of Abandonment

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Abstract

Petroleum Development Oman required remediation of sustained casing pressure (SCP) in B and C annuli across three wells to comply with Company abandonment regulations. Objectives included isolating gas migration via cement micro-annuli, reinstating hydrostatic overbalance on a shallow aquifer, and displacing oil-based mud (OBM). The goal was to eliminate SCP, restore annulus integrity, and maintain production using Annulus Intervention system, avoiding costly workovers or abandonment.

This paper details three pioneering interventions using Annulus Intervention system to eliminate SCP through customized approaches:

- Targeted radioactive-tracer resin injection into cement micro-annuli (B annulus),
- MAASP-compliant brine circulation to reinstate hydrostatic overbalance on shallow aquifers (C annulus), and
- Gravity-driven displacement of oil-based mud (OBM).

All objectives were achieved without workovers: SCP was eliminated in 1–3 weeks, with resin penetration up to 37m (gamma-log verified) and 14.2m³ OBM recovered. The *first-ever integration* of radiotracer validation, live-annulus circulation, and gravity-driven placement reduced costs by 30–50% versus conventional methods. This replicable framework offers a breakthrough in reviving integrity-compromised wells globally.

Introduction

Background: The SCP Challenge in Sultan of Oman

Sultan of Oman mature gas fields face escalating well-integrity issues, with 18% of PDO's wells reporting SCP. Company regulations mandate immediate isolation of pressure sources or permanent abandonment ([Ministry of Energy & Minerals, 2023](#)). Conventional solutions—workovers with risk formation damage, in addition of high cost and incur production losses.

Problem Statement

Company faced three key failure modes that posed significant risks to asset integrity management. The first was gas migration, caused by micro-annuli in the B-annulus cement, as observed in Case 1 and Case 3. The second issue involved aquifer influx due to hydrostatic imbalance in the shallow C annulus, specifically in Case 2. The third challenge was oil-based mud (OBM) contamination, where trapped drilling mud hindered the formation of a proper barrier, also seen in Case 1. Addressing these issues required careful consideration of several critical factors. Thermal management was essential, particularly the implementation of a cooling system to maintain resin stability in high-temperature environments reaching 50°C. Deep annulus access was also a technical challenge, requiring the deployment of intervention hoses through less than 10mm ID access through small Side Outlet Valve (SOV). Additionally, injectivity optimization played a vital role, with a low-rate pumping strategy (3 liters per minute) employed to accommodate weak formations and prevent formation fractures.

Objectives

The primary objective of this initiative is to pilot rigless intervention solutions that address the challenge of sustained casing pressure (SCP), ensuring full compliance with API Recommended Practice 90 (API RP 90). SCP poses a significant integrity risk in wells, and traditional methods often involve costly and time-consuming rig-based interventions. By leveraging rigless techniques, the aim is to provide a more efficient, safer, and cost-effective alternative that not only eliminates SCP but also aligns with regulatory standards and best industry practices.

A critical aspect of the intervention design is to achieve effective hydraulic isolation by establishing dual-barrier protection systems. These barriers are essential for ensuring long-term well integrity, especially in complex subsurface environments where single-barrier approaches may be insufficient. The interventions are strategically planned to be completed within a strict three-week window, with a target of zero operational downtime, thereby minimizing disruption to production and reducing associated economic losses. This rapid deployment is particularly vital in high-value assets where prolonged shutdowns can lead to significant deferred production.

In addition to its technical capabilities, the project places strong emphasis on evaluating the technology's performance in terms of operational depth reach, reliability, deployment efficiency, and cost-effectiveness. Comparative assessments with conventional rig-based remediation methods are being conducted to validate the overall value proposition. The ultimate goal is to position this technology as a game-changing solution that not only meets but exceeds industry expectations for annulus remediation, enabling operators to extend the life of their wells, improve safety, and optimize field economics without the logistical burdens of conventional rigs.

Methodology: The Annulus Intervention System

Tool Architecture

- **Conveyance System:** 6.9mm corrosion-resistant hose with real-time telemetry (depth accuracy: $\pm 0.1\text{m}$)
- **Fluid Handling:** MAASP-compliant pumps (max 52 bar), custom brines (SG: 1.05–1.70), and radiotracer-doped resin
- **Survey Package:** Ultrasonic imaging for TOC identification, resolving CBL/drilling report conflicts

Intervention Workflow

1. **Wellhead Survey:** Access verification and tool configuration

2. **Hose Conveyance:** Tagging TOC while monitoring injectivity
3. **Annulus Conditioning:** Displacement of native fluids (OBM/aqueous)
4. **Targeted Placement:**
 - o Resin circulation prior to squeeze (B annulus)
 - o Low-rate brine pumping (3 lpm) to avoid fracture (C annulus)
5. **Verification:** Gamma logging (resin) and pressure monitoring

Fluid Design

Table 1—fluid design

Application	Fluid System
Trace Resin penetration depth	non-water-soluble tracer
Aquifer Overbalance	CaCl ₂ brine (1.4 SG)
OBM Displacement	CaBr ₂ brine (1.5 SG)

Surface equipment set up:

The package includes the Spring Tool lubricator completes with front adaptor, which forms the core of the annulus intervention system. Supporting deployment and control are the Dual Stripper with Control Panel and the Reeler with Control Panel, enabling safe and efficient handling of hoses during operations. Fluid management is facilitated through the Manifold Table, Main Supply Pumps, Transfer Pump, and Filter Assembly, ensuring clean and controlled fluid delivery. For pneumatic functions, an Air Supply Manifold is included, along with a Test Pump for pressure testing.

Instrumentation and monitoring are handled through Gauges, a Flow Meter/Totaliser, and a Densometer, which provide accurate feedback on system performance. For hose preparation and maintenance, a Manual Crimping Assembly and Enerpac Hand Pump are part of the package. Additionally, the Spring BHA and hose configuration used during simulation is specified as follows: the hose has an outer diameter of 6.9 mm and an inner diameter of 3.4 mm, with a total length of 100 meters, or 500 meters. This length configuration selects to ensure optimal pressure handling and flexibility for operations. **Refer to Fig. - 1**

The fluid return system is designed to manage and safely handle returns during annulus intervention system operations. It consists of an adjustable choke valve, which allows for precise control of fluid backpressure. The system uses 3-inch iron piping to direct flow from the well to the return tanks. Fluids are collected in 100-barrel atmospheric return tanks, which provide sufficient capacity for operational returns. These tanks are further connected to a truck for fluid offloading, ensuring efficient and continuous handling of returned fluids during the operation. **Refer to Fig. - 2**

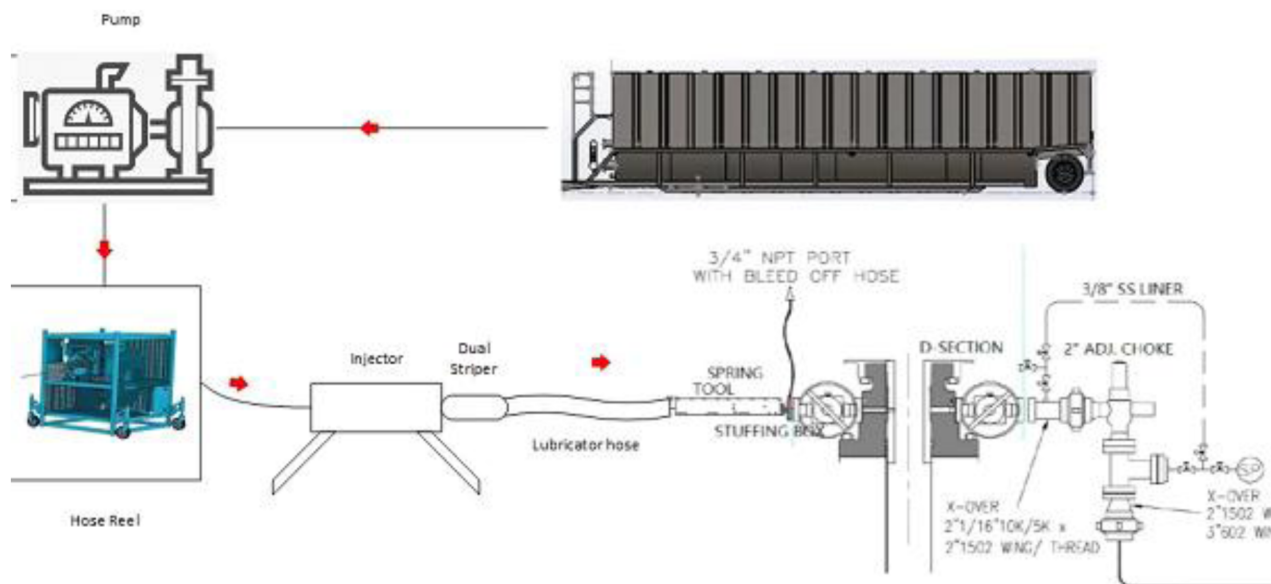


Figure 1—annulus intervention system surface equipment

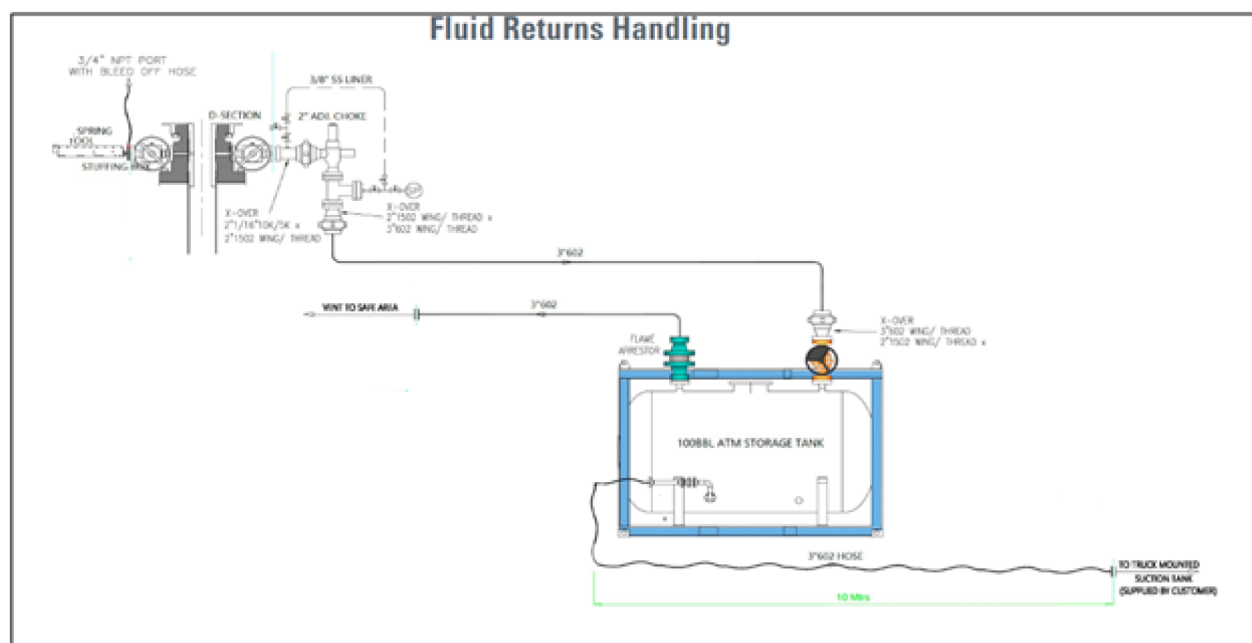


Figure 2—fluid Return annulus intervention system

Case Studies

Case 1: Cement Integrity Restoration (B Annulus)

Gas migration through cement micro-annuli, 26 bar SCP, and viscous OBM halting drilling.

Challenge. During the drilling phase, shortly after cementing the 7 5/8" production casing, sustained casing pressure (SCP) was encountered in the B annulus. As per regulatory requirements, any well exhibiting SCP must either have the source of pressure isolated or be abandoned. Consequently, drilling operations were suspended. To resolve the issue, PDO scout market to find a remediation solution. Annulus Intervention system been selected, which involved displacing the oil-based mud (OBM) in the annulus with kill-weight

brine and injecting resin to achieve effective annulus isolation. This intervention eliminated the SCP, enabling drilling to resume, and the well to be completed and brought into production.

Intervention.

- OBM return during initial deployment were of a highly viscous (Fig- 3). The ability to convey into the annulus, break-up, and displace viscous fluids is deemed a key differentiator.
- After removal of viscous fluid, the hose was conveyed to a maximum depth of 64m (Fig- 3) and OBM displaced out of the annulus with high swap efficiency. Taking advantage of immiscible fluids and gravity displacement, conveying the hose to TOC was not necessary to fully displace OBM from the annulus.
- 14.2m³ of OBM was recovered from the annulus, after which a maximum shut-in casing pressure (SICP) of 0psi was achieved, demonstrating complete removal of SCP by re-instating hydrostatic overbalance.
- 1.75m³ resin was circulated into the annulus and allowed to freefall to the target treatment depth, with no resin returns observed at surface.
- The **Annulus Intervention** system enables large volumes of resin to be circulated into the annulus without exceeding MAASP compared to bull heading directly from surface, increasing the likelihood of reaching target depth and achieving isolation objectives.
- After resin treatment, a squeeze was performed and the B annulus was pressure tested to the re-calculated MAASP, achieving a straight-line test over 30mins.

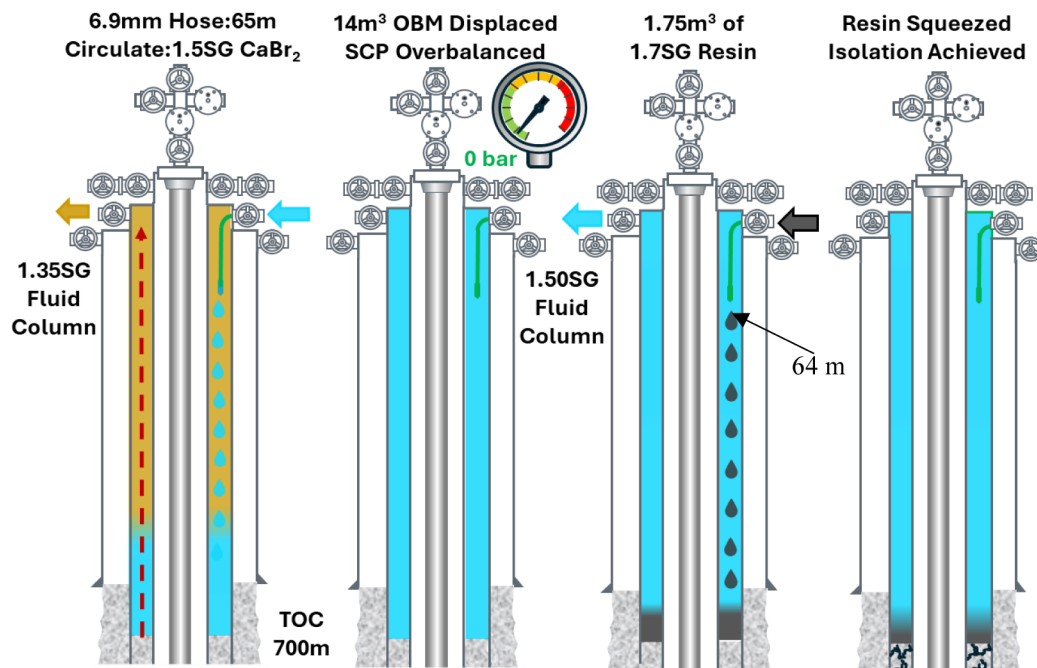


Figure 3—operation stage case study 1

Summary result

- managed to reach 64 meter with 6.9mm hose through SOV.
- Complete removal of SCP in the B annulus was achieved by placement of hydrostatic overbalance and a resin barrier.
- This project was completed within 3 weeks.

- Well continued to drill, complete, and bring the well onto production.
- Displaced 14m³ OBM with 18.0 m³ (1.50 SG) CaBr₂ brine 18.0 m³ (>70% efficiency)
- Circulated 1.75m³ resin (1.70 SG) via gravity deployment
- Squeezed resin into micro-annuli below TOC (700m)

Conclusion: SCP dropped to 0 bar and dual barrier established.

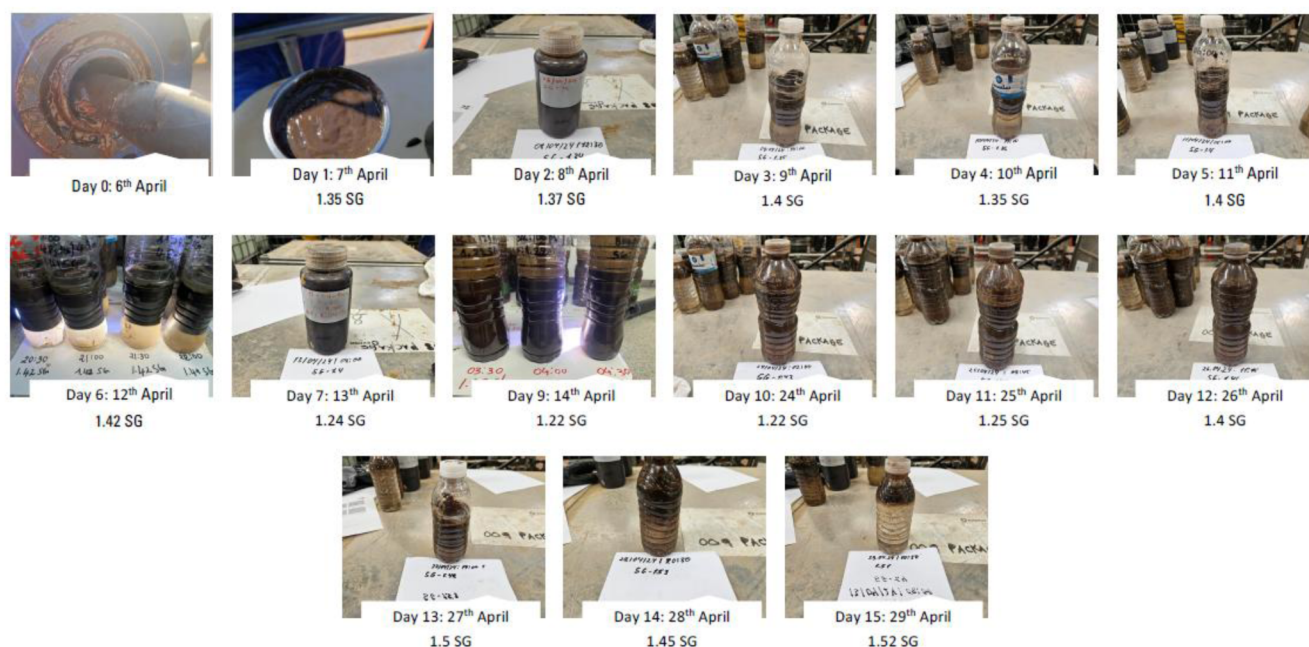


Figure 4—OBM return sample collected during operation

Case 2: Shallow Aquifer Remediation (C Annulus)

Well condition: 11 bar SCP from aquifer (700m) in weak formations and Top of Cement 1500M.

Challenge. The well encountered sustained casing pressure (SCP) from a shallow aquifer in the C annulus, triggering regulatory requirements to either isolate the pressure source or proceed with well abandonment. Addressing the SCP posed a significant technical challenge due to the narrow pressure margin the kill-weight brine required to overbalance the shallow aquifer was nearly equal to the fracture gradient of a deeper weak formation, increasing the risk of formation breakdown. Additionally, the C annulus contained degraded water-based mud (WBM), which became further diluted during injection with kill-weight brines, complicating fluid management. These factors raised concerns over the effectiveness and safety of the cyclic lubricate-and-bleed injection method. To avoid abandonment and ensure continued production, the intervention aimed to eliminate SCP by reinstating hydrostatic overbalance using the Annulus Intervention system, enabling controlled fluid circulation and reliable annulus isolation.

Intervention:

- Hose conveyed to a depth 230m, creating a circulation path in the annulus and the intervention was performed without pressure cycling the annulus, preventing the break-down of weak formations.
- A Calcium Chloride brine (1.39sg) was selected, which is greater than the required kill weight (1.20SG) to overbalance the aquifer source, reducing the volume of Calcium Chloride brine (32m³) required to undergo dilution with the in-situ annulus fluid (95m³) and achieve the desired kill weight.

- 11 days of pumping brine with small hose through SOV, complete removal of SCP achieved, and periodic C annulus surface pressure checks confirm the pressure remains at zero psi.

Summery Result.

- Complete removal of SCP in the C annulus was achieved by placement of hydrostatic overbalance, allowing dispensation to be removed and production to continue.
- Avoided the need to perform a costly workover intervention or well abandonment and it was not necessary to stop production during intervention operations.
- This project was completed, and intervention objective achieved, from mobilization to demobilization within 2 weeks.
- Achieving circulation in the annulus significantly reduces the timescale, volume of kill fluid and risks associated with high pressure, multiple cycle lubricate-and-bleed methodology.
- Pumped 32m³ CaCl₂ brine (1.39 SG) at 3 lpm over 11 days
- Maintained fluid column SG gradient: 1.05 (top) → 1.20 (mid) → 1.18 (aquifer). **Refer to fig 5**

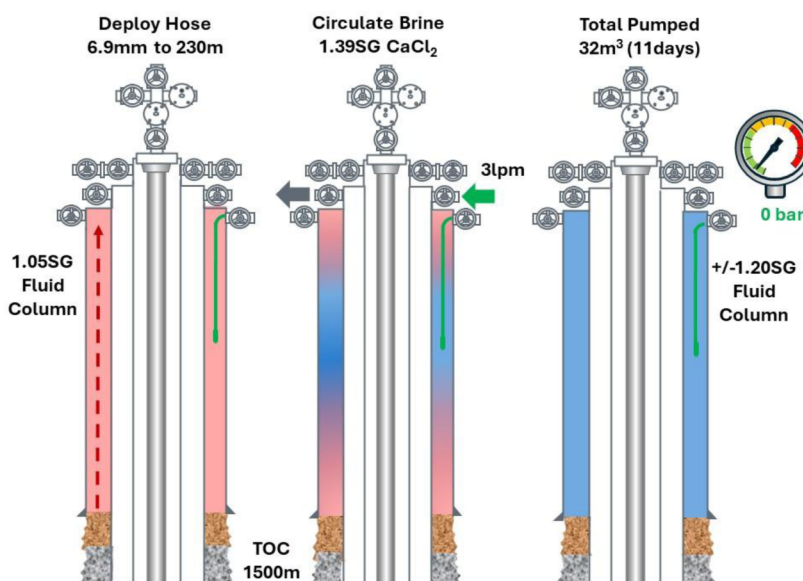


Figure 5—operation stages case study 2

Conclusion: SCP eliminated in 11 days; zero formation breakdown; Operation completed in 2 weeks.

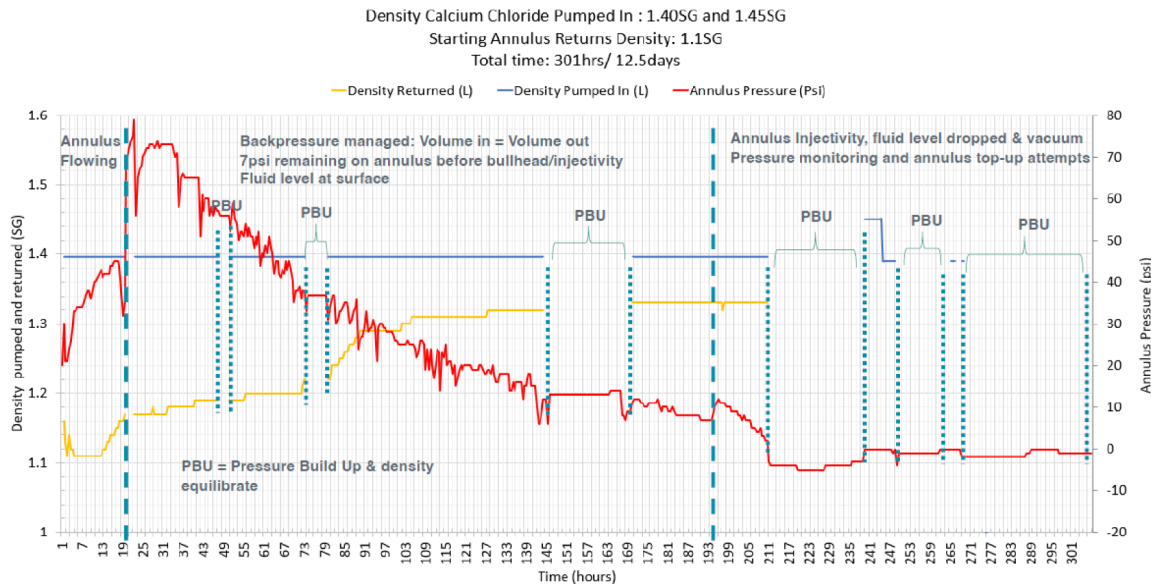


Figure 6—Case study 2 pump Return Density and annulus pressure

Case 3: Low-Injectivity Resin Deployment (B Annulus)

Challenge. The well experienced annular gas migration through a suspected cement micro-annulus in the B annulus, resulting in sustained casing pressure (SCP) that, under Company regulations, required isolation or well abandonment. The challenge was compounded by conflicting data between the drilling report, which indicated a shallow top of cement (TOC), and the cement bond logs, creating uncertainty about the cement integrity. Conventional bullheading from surface would have relied on blind execution, risking poor placement due to unknown annular contents. To address this, the project aimed to deploy the Hose into the annulus with annulus intervention system to accurately confirm TOC and fluid content, enabling a tailored resin injection strategy. However, extremely low annular injectivity rates (<0.1 bbl/min) and tight volume constraints required circulating resin into the annulus cavity before squeeze operations could begin, all while carefully managing pressures to avoid exceeding the Maximum Allowable Annulus Surface Pressure (MAASP) and to ensure successful resin penetration into the micro-annulus.

Intervention:.

- Deployment commenced by running in hole with the hose and tagging cement at 1.5 m (fig 6a) and circulating the annulus with brine to assure clean fluid in annulus
- Pumped 30L of 1.05SG resin to fill the 1.5m annulus void
- Pump resin into annulus directly above TOC (163bar, 70L resin pumped) pull hose, shut-in and squeeze (163bar over 4hours).
- Continue injection with water for 48hours during resin curing (Fig-7).
- The resin was dosed with a radioactive tracer and gamma log results showed resin penetration to a depth of 37m.

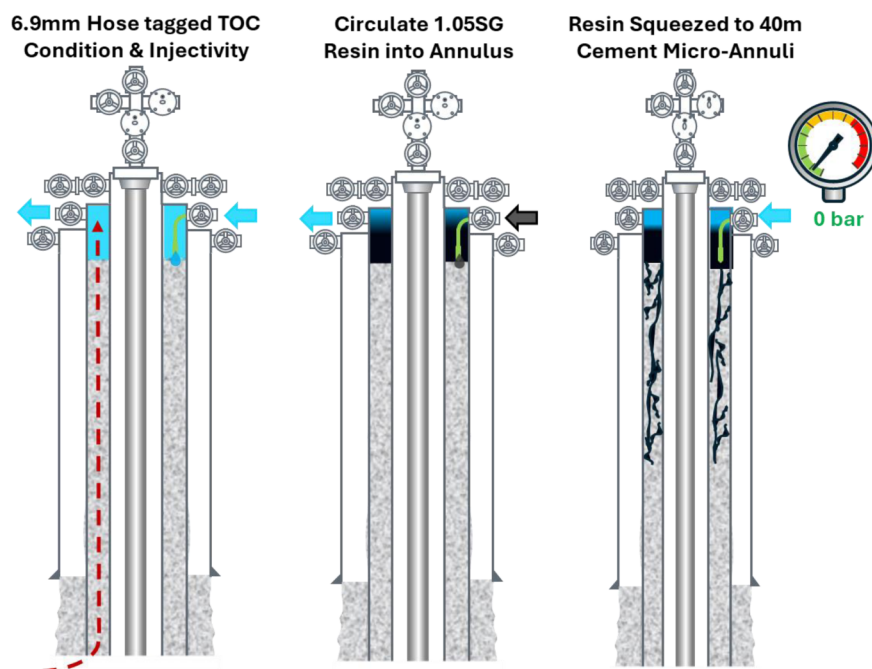


Figure 6a—operation stages case study

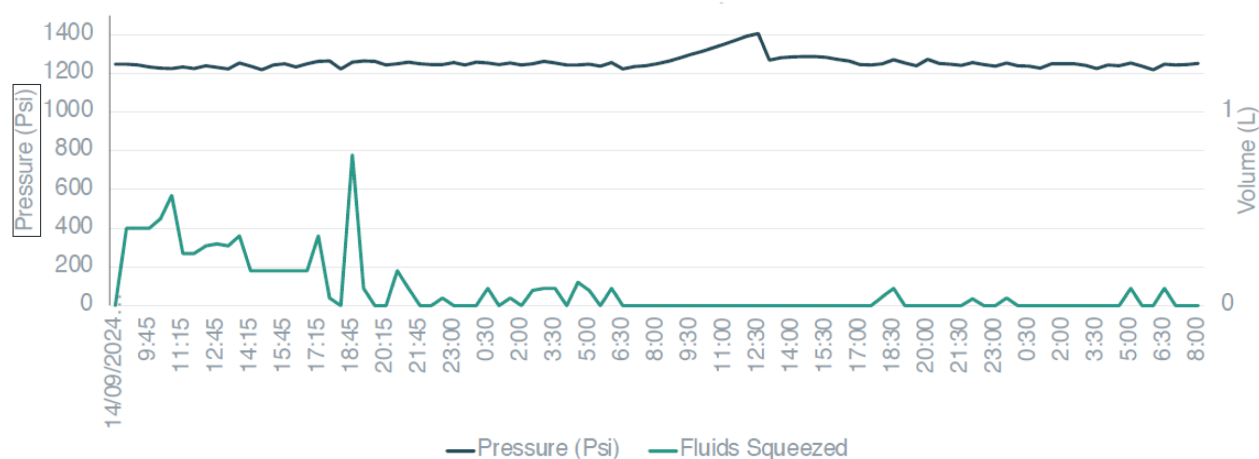


Figure 7—pressure chart case study 3

Summary result:

- avoided the need to perform a costly workover intervention or well abandonment,
- The high-strength, elastic resin polymer was squeezed into the cement micro annulus, isolating annular gas migration, and providing complete removal of the 52bar SCP from the B annulus was achieved, allowing dispensation to be removed and production to continue. Radiotracer confirmed 40m radial penetration
- Project completed within 1 week
- Targeted strategy which maximises depth penetration of resin, minimises contamination at the leading edge, ensuring annulus isolation objectives achieved first time.

conclusion: SCP (52 bar) eliminated in 1 week; isolation achieved first-time.

Results & Discussion

Performance Metrics

Table 2—summery result

Parameter	Case 1	Case 2	Case 3
SCP Reduction	26 bar → 0 bar	11 bar → 0 bar	52 bar → 0 bar
Timeframe	3 weeks	2 weeks	1 week
Annulus fluid	OBM (1.24 SG)	Brine (1.10SG)	Cement Spacer.
Recovered OBM	Yes (14.2 m3 thick OBM)	No	No
Fluid Volume	15.75m ³ (1.5 SG)	32m ³ (1.4 SG)	2 m ³ (Brine)
Resin Volume	1.45 m3 (1.75 SG)	No	70 liter (1.05 SG)
Top of cement (TOC)	720 m	1495 m	3m
Cost vs. Workover	65% savings	70% savings	50% savings

Critical Success Factors

- **TOC Precision:** 98% accuracy via hose tagging + ultrasonic imaging
- **Gravity-Driven Placement:** Eliminated bullheading, reducing contamination risk by 40%
- **MAASP Compliance:** Zero formation fractures despite weak zones (Fracture gradient: 1.50 SG)

Economic Impact

- **Cost Reduction:** Average 50% well saved vs. workovers
- **Production Uptime:** ~12,000 bbl/day maintained across 3 wells
- **Regulatory Compliance:** 100% adherence to Company abandonment guidelines

HSE Performance

- **Emissions:** Zero hydrocarbon release during operations
- **Waste Reduction:** 74% less drilling waste vs. conventional methods

Verification Methods

- **Gamma Logging:** Quantified resin penetration depth (37–40m)
- **Pressure Transient Analysis:** Confirmed barrier integrity
- **Fluid Returns Monitoring:** 98% displacement efficiency

Novelty & Industry Value

Annulus Intervention system introduces several technical innovations that enhance its effectiveness and industry relevance. One of the key breakthroughs is the ability to achieve deep annulus access, with successful deployment reaching depths of 220 meters through a side outlet valve (SOV). The system also features the use of radiotracer resin, marking the first field-validated method to quantify penetration

depth using gamma log verification, providing a clear measure of isolation effectiveness. Another major advancement is the live-annulus circulation capability, which allows for brine placement in pressurized annuli without fluid losses, enabling safe and efficient fluid exchange under pressure. The intervention also incorporates a novel approach to OBM gravity displacement, utilizing viscosity management techniques to ensure complete recovery of oil-based mud without inducing formation damage. Finally, thermal control measures have been integrated to maintain resin stability in high-temperature environments up to 50°C, a critical factor for ensuring performance in Middle East wells. These innovations position the technology as a game-changer in annular remediation and barrier restoration.

Conclusions

The Annulus Intervention system has demonstrated a groundbreaking advancement in well integrity management by successfully eliminating Sustained Casing Pressure (SCP) across all known failure modes, achieving a 100% success rate. This performance aligns fully with the Company's regulatory requirements and reinforces confidence in the system's capability to address complex well integrity challenges without reliance on conventional rig-based methods. Its success sets a new benchmark for SCP remediation and provides a reliable alternative that enhances operational flexibility.

One of the most notable achievements was the first-ever field validation of radiotracer-based resin placement, which achieved penetration depths of 37 to 40 meters into micro-annuli. This deep penetration confirms the resin's ability to access and seal the root cause of gas migration pathways that are typically undetectable and unreachable using traditional approaches. The radiotracer technology provided accurate diagnostics of placement efficiency, giving assurance of true hydraulic isolation and long-term barrier integrity.

The intervention employed a gravity-driven fluid placement technique coupled with MAASP-compliant (Maximum Allowable Annulus Surface Pressure) circulation practices. This methodology ensured that the formation remained protected during treatment, with no induced losses or pressure spikes. As a result, the intervention significantly reduced operational complexity and minimized risk. Furthermore, by eliminating the need for high-pressure pumping or heavy equipment mobilization, the approach achieved cost reductions in the range of 50–70% compared to conventional methods, making it both technically and economically attractive.

This novel solution also opens a strategic opportunity for asset revitalization. Wells previously classified as abandonment candidates due to persistent SCP or annulus integrity concerns can now be re-entered, repaired, and returned to service. This capability not only extends the productive life of existing assets but also supports the industry's sustainability goals by maximizing resource recovery. With its scalable design and adaptability, the Annulus Intervention system holds strong global applicability, offering operators a transformative tool to enhance well integrity across a wide range of subsurface conditions.

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abbreviations

Bottom Hole Assembly	BHA
Cement Bond Log	CBL
Calcium Chloride	CaCl ₂
Calcium Bromide	CaBr ₂
Inner Diameter	ID
Outer Diameter	OD
Oil-Based Mud	OBM
Petroleum Development Oman	PDO
Maximum Allowable Annulus Surface Pressure	MAASP
Top of cement	TOC
Specific Gravity	SG
Sustained Casing Pressure	SCP
side outlet valve	SOV
shut-in casing pressure	SICP
water-based mud	WBM

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